Recurrence Equations

Recurrence Equations, Solving Recurrence Equations- Substitution method and Iteration method

Recurrence relations

- Recurrence is an equation or inequality that describes a function in terms of its value on smaller inputs, and one or more base cases
- e.g., recurrence for Merge-Sort

$$T(n) = \begin{cases} \Theta(1) & \text{if } n = 1 \\ 2T(n/2) + \Theta(n) & \text{if } n > 1 \end{cases}$$

- Useful for analyzing recurrent algorithms
- Make it easier to compare the complexity of two algorithms
- Methods for solving recurrences

- Substitution method
- Recursion tree method
- Master method
- Iteration method

Iteration method

to expand the recurrence and express it as a summation of terms of n and initial condition.

Example1: Consider the Recurrence

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T (n) = 1  if n=1
      = 2T (n-1) if n>1
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Solution:

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T (n) = 2T (n-1)
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Repeat the procedure for i times

$$T(n) = 2^i T(n-i)$$

Put $n-i=1$ or $i= n-1$ in (Eq.1)

$$T(n) = 2^{n-1} T(1)$$

$$= 2^{n-1} .1 \quad \{T(1) = 1 \text{ .....given}\}$$

$$= 2^{n-1}$$

Example2: Consider the Recurrence

$T(n) = T(n-1) + 1$ and $T(1) = \theta(1)$.

$$
\begin{aligned}
T(n) &= T(n-1) + 1 \\
&= (T(n-2) + 1) + 1 = (T(n-3) + 1) + 1 + 1 \\
&= T(n-4) + 4 = T(n-5) + 1 + 4 \\
&= T(n-5) + 5 = T(n-k) + k
\end{aligned}
$$

Where $k = n-1$

$T(n-k) = T(1) = \theta(1)$

$T(n) = \theta(1) + (n-1) = 1 + n - 1 = n = \theta(n)$.

Substitution Method

- ★ Guess the Solution.
- ★ Use the mathematical induction to find the boundary condition and shows that the guess is correct.

For Example1 Solve the equation by Substitution Method.

$$T(n) = T\left(\frac{n}{2}\right) + n$$

We have to show that it is asymptotically bound by O (log n).

Solution:

For $T(n) = O(\log n)$

We have to show that for some constant c

$T(n) \leq c \log n.$

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Put this in given Recurrence Equation.

$$T(n) \leq c \log\left(\frac{n}{2}\right) + 1$$

$$\leq c \log\left(\frac{n}{2}\right) + 1 = c \log n - c \log_2 2 + 1$$

$$\leq c \log n \text{ for } c \geq 1$$

Thus **$T(n) = O \log n$.**

$$T(n) = 2T\left(\frac{n}{2}\right) + n \quad n>1$$

Find an Asymptotic bound on T.

Solution:

We guess the solution is O (n (logn)).Thus for constant 'c'.

$T(n) \leq c\ n\ \log n$

Put this in given Recurrence Equation.

Now,

$$T(n) \leq 2c\left(\frac{n}{2}\right)\log\left(\frac{n}{2}\right)+n$$

$$\leq cn\log n - cn\log 2 + n$$

$$= cn\log n - n(c\log 2 - 1)$$

$$\leq cn\log n \text{ for } (c \geq 1)$$

Thus **T (n) = O (n logn).**